

TrES-3b

R-2366

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120415 · created 2026-08-19T23:55:10+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456032.85802 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.012
Transit depth (Rp/Rs)^2	1.64 +/- 0.31 [%]
Orbital Inclination (inc)	83.9 +/- 0.72
Airmass coefficient 1 (a1)	0.9821 +/- 0.0098
Airmass coefficient 2 (a2)	0.0127 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#3 - [391, 387]
Optimal Method	PSF photometry
Transit Duration (day)	0.0663 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.128 ± 0.012 vs 0.16309 (deviation 2.47σ)
Duration fit vs published	0.0663 d vs 0.0567 d (deviation 1.85σ)
Mid-time O-C	-1.4 min
Depth SNR	9.0
Residual scatter	0.99 %

Light curve

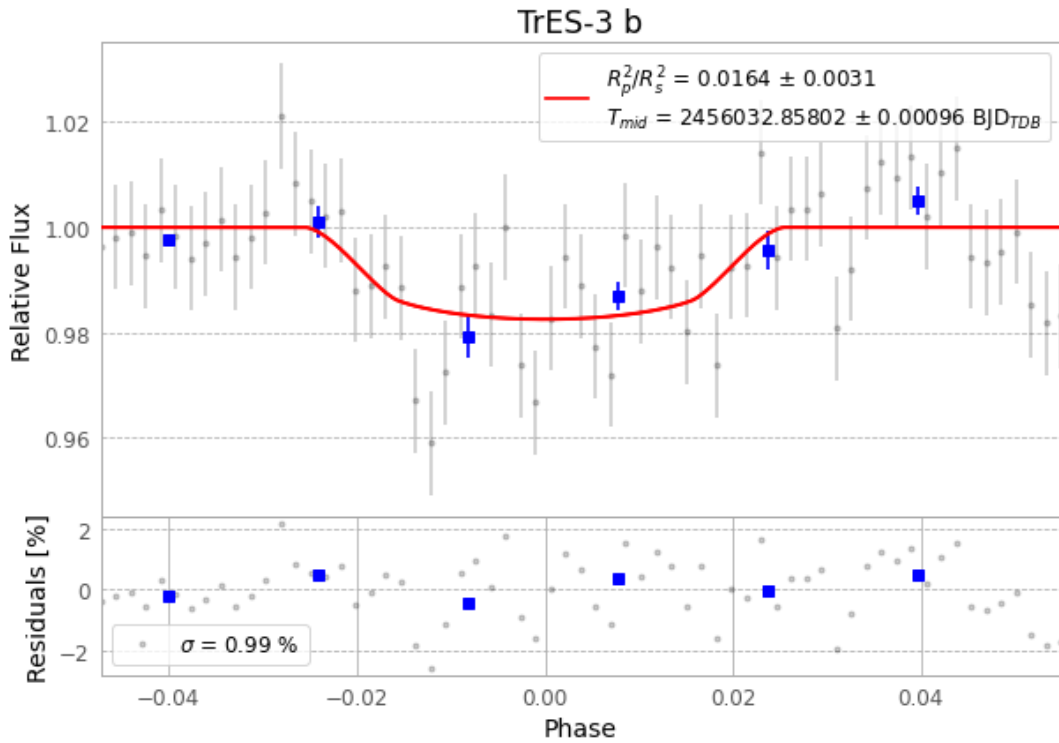


Figure 1 — FinalLightCurve_TrES-3 b_2012-04-15.png (SHA-256 e12a89a02d1b7a0e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [343, 176], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c2be6a5ad66fe2ec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3120415070312.FITS → TRES-3120415101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120415.log (32ec2dd16d5b...), FinalLightCurve_TrES-3 b_2012-04-15.png (e12a89a02d1b...), FinalLightCurve_TrES-3 b_2012-04-15.csv (06f594173498...)

Compute resources

Wall time 400.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:55Z.